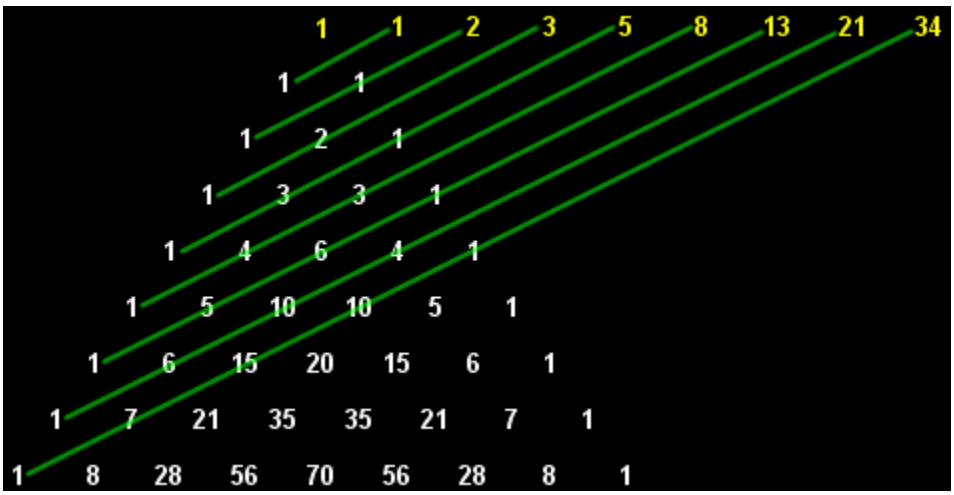
American Computer Science League

2021-2022 ● Contest 3: Fibonacci & Pascal ● Senior Division

PROBLEM: One of the many patterns that can be found in Pascal's Triangle is where the Fibonacci sequence 1, 1, 2, 3, 5, 8, 13, 21, 34, … can be found by adding the numbers on the diagonals as illustrated with the following diagram:



In this program, generate each row of Pascal's Triangle by placing a 1 in the first and last position. For the positions in the middle of the triangle, every number is the sum of the numbers immediately above it. The 0th row contains a single 1. For row N, there are N+1 items.

INPUT: There are 5 lines of data. Each line has a single integer representing one of the Fibonacci numbers in the following sequence: 1 1 2 3 5 8 13 21 34 55 89 .... We guarantee it will be less than 2^{63} since that is the largest integer that can be represented using 64 bits.

OUTPUT: For each line of data, find all of the numbers on that diagonal used to get that sum and any of the previous sums in the triangle. Find all of the integer(s) that occur only once and print how many there are. For example, in the triangle shown, if the input is 8, then the numbers found in this sum and any of the previous ones are {1, 4, 3} for 8, {1, 3, 1} for 5, {1, 2} for 3, {1, 1} for 2, and {1} for 1. The 2 and the 4 occur only once, so 2 is printed.

SAMPLE INPUT:

```
8
89
610
10946
317811
```

SAMPLE OUTPUT:

```
1. 2
2. 8
3. 16
4. 31
5. 58
```

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TEST DATA

TEST INPUT:

```
55
1597
832040
9227465
1836311903
```

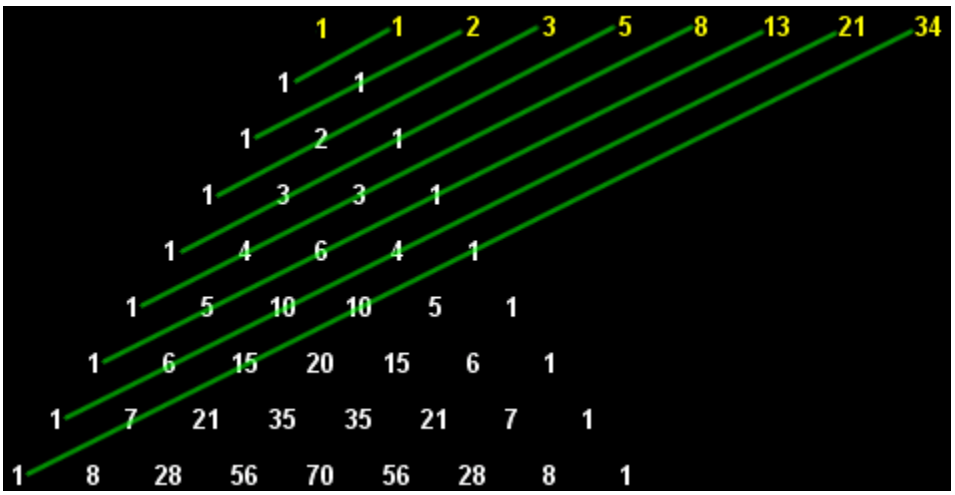
TEST OUTPUT:

1. 6
2. 21
3. 67
4. 96
5. 171

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题目: 在帕斯卡三角形中能够发现一些性质，如通过将帕斯卡三角形对角线上的数字相加，便可以找到斐波那契数列 1, 1, 2, 3, 5, 8, 13, 21, 34, … 如下图所示：



在上图中，帕斯卡三角形的每一行的首尾位置都是 1。在三角形中间位置上的每个数字都是其上方紧邻的两个数字之和 。第 0 行只有一个数字 1，第 N 行有 N+1 个数字。

输入: 输入 5 行数据。每行都有一个整数，表示斐波那契数列中的某个元素：1 1 2 3 5 8 13 21 34 55 89 .... 我们保证所有数字小于 2^{63} ， 因为 2^{63}是可以用 64 位表示的最大整数。

输出: 根据输入的每行数据，找出三角形对角线上数字和等于给定整数的所有数字，以及对角线上方出现的所有数字。找出只出现过一次的所有数字，并输出这些数字的总数量。例如，在所示的三角形中，如果输入的数据是 8，找到对角线上相加为 8 的所有数字，以及对角线上方的所有数字，则有 {1, 4, 3} 等于 8，{1, 3, 1} 等于 5，{1, 2} 等于 3，{1, 1} 等于 2，及 {1} 等于1。只有数字 2 和数字 4 两个数字出现了一次，所以输出 2 。

示例输入:

```
8
89
610
10946
317811
```

示例输出:

```
1. 2
2. 8
3. 16
4. 31
5. 58
```

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测试数据

测试输入：

```
55
1597
832040
9227465
1836311903
```

测试输出：

1. 6
2. 21
3. 67
4. 96
5. 171